

SOC Analyst Home Lab Project using Splunk Cloud

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1. Project Overview

This project demonstrates a basic SOC Analyst workflow using Splunk Cloud to monitor failed login attempts, analyze authentication logs, and configure alerts for suspicious activity.

The goal of this project was to gain hands-on experience with SIEM tools, log analysis, dashboard creation, and alert configuration.

2. Objectives

- Detect failed login attempts
- Analyze authentication logs
- Create SPL queries
- Build dashboards for visualization
- Configure alerts for suspicious activity

3. Tools & Technologies Used

- Splunk Cloud
- SPL (Search Processing Language)
- SIEM Monitoring
- Log Analysis
- Dashboard Creation
- Alert Configuration

4. SPL Query Used

```
index=main  
| search status=failed  
| stats count by username  
| sort - count
```

Explanation:

- Searches failed authentication events
- Counts failed attempts by username
- Sorts results from highest to lowest

5. Query Results

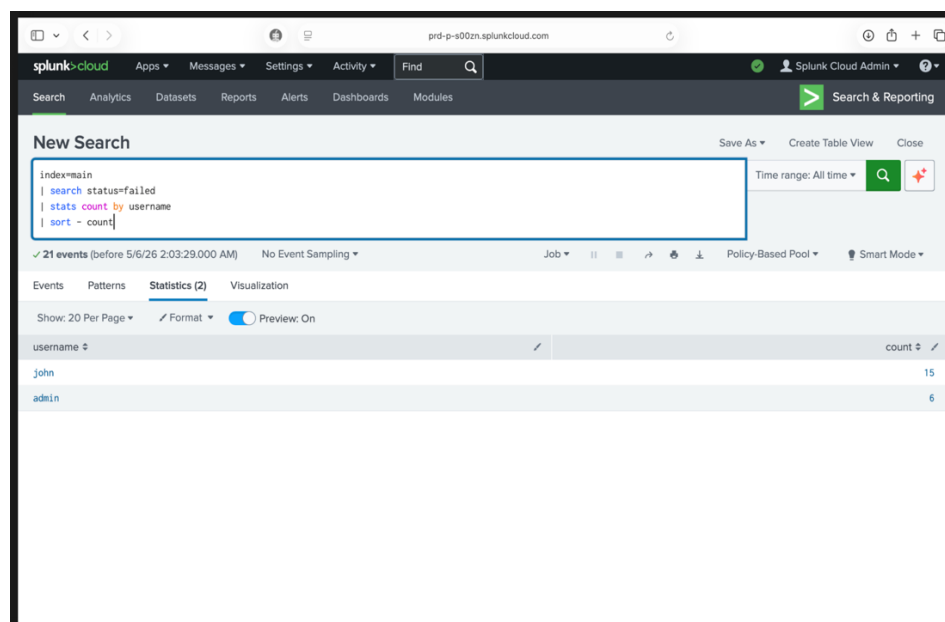


Figure 1: SPL query showing failed login attempts by username.

6. Dashboard Visualization

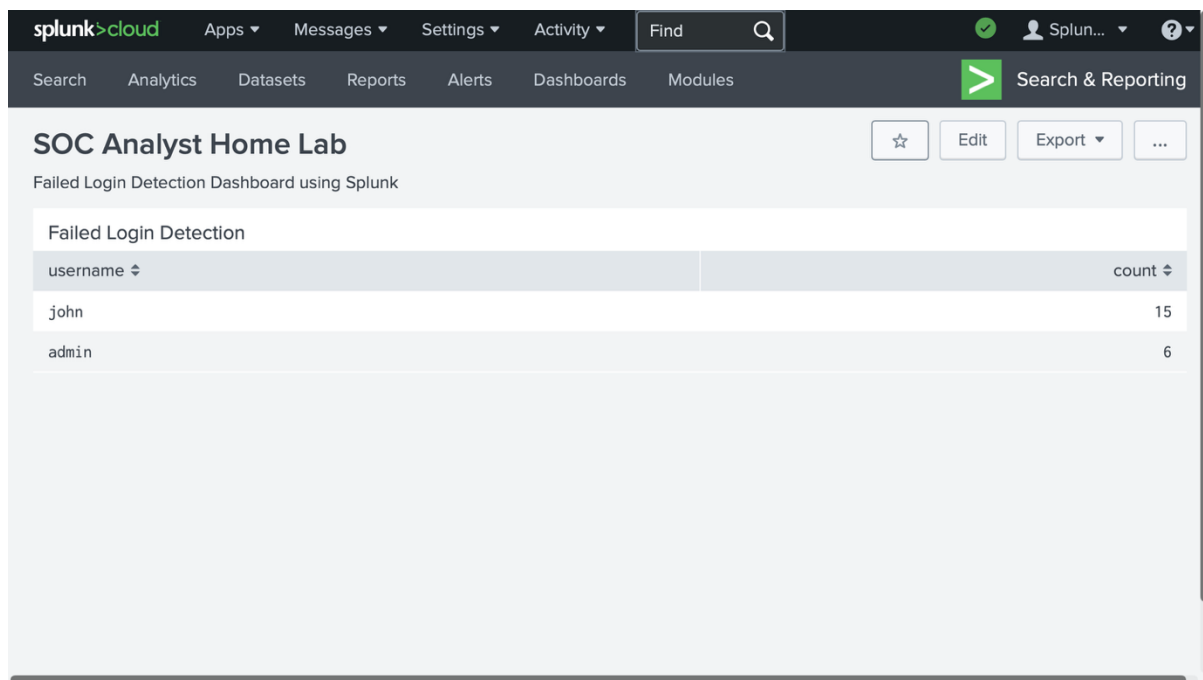


Figure 2: Failed Login Detection Dashboard created in Splunk Cloud.

7. Alert Configuration

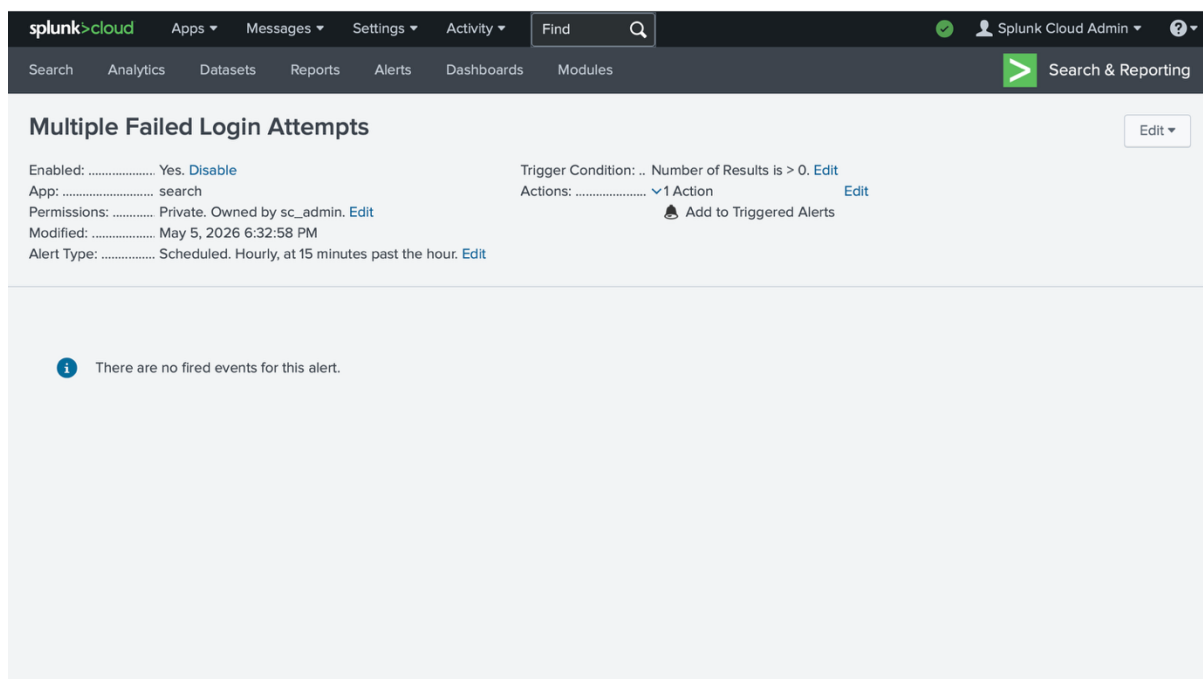


Figure 3: Scheduled alert configured for detecting multiple failed login attempts.

8. Key Learnings

Through this project, I learned:

- Basics of SIEM monitoring
- Log analysis using Splunk
- SPL query creation
- Dashboard visualization
- Security alert configuration
- Threat detection workflow

9. Conclusion

This project helped me understand how SOC analysts monitor suspicious authentication activity and respond to potential brute-force login attempts using SIEM tools like Splunk.